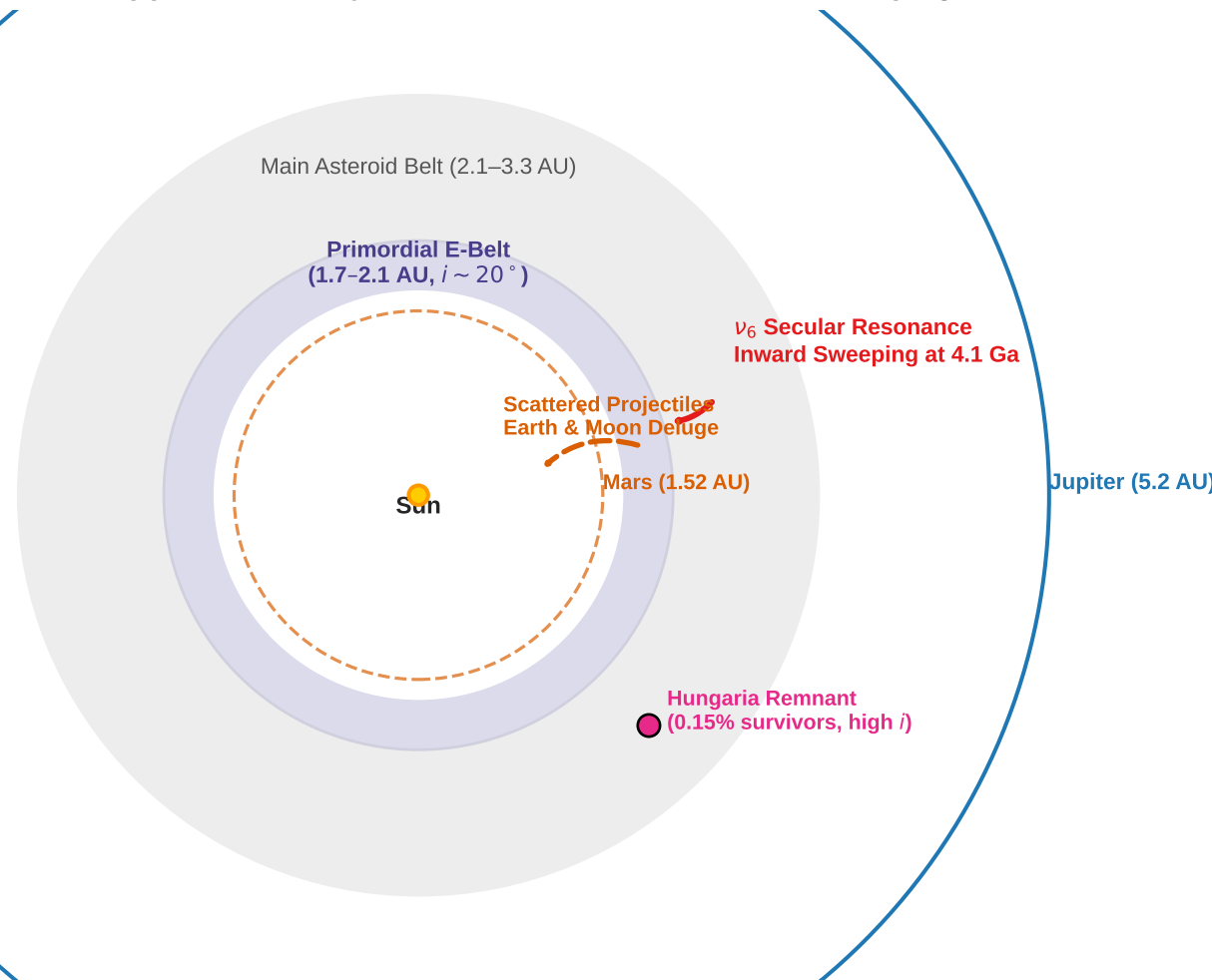


(b) E-Belt Breakdown Dynamical & Geological Timeline

(a) Inner Solar System Architecture & Resonance Sweeping



1. Giant Planet Orbital Instability (~ 4.1 Ga)
 - Jupiter/Saturn resonance crossing
 - Inward migration of Saturn g_6 frequency



2. E-Belt Destabilization & Prompt Pulse (4.1–3.8 Ga)
 - ν_6 secular resonance sweeps $1.70 \leq a \leq 2.10$ AU
 - $\sim 82\%$ of E-belt bodies ejected on $\tau_1 \approx 35$ Myr
 - Creates Nectaris, Imbrium, Orientale Lunar Basins



3. Archaean Impact Deluge (3.8–1.7 Ga)
 - Slow chaotic/Yarkovsky feeding tail ($\tau_2 \approx 440$ Myr)
 - Delivers ~ 15.5 giant impacts ($D \geq 180$ km) to Earth
 - Form documented Barberton & Pilbara spherule layers



4. Modern Epoch (Present Day)
 - Surviving 0.15% forms high-inclination Hungaria family
 - $a \approx 1.90$ AU, $e \approx 0.08$, $i \approx 22^\circ$
 - Reconciles lunar basins and Archaean terrestrial spherules